Geodesics and Gravitational Redshift as Projection Necessities in the Methane Metauniverse

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Abstract

We show that Einstein gravity in the weak-field limit is an unavoidable consequence of the Methane Metauniverse (MMU) framework. Starting from a discrete elastic lattice of ur–cells with internal projection-defined time, we derive the Poisson field equation, the Schwarzschild redshift factor, and geodesic motion without assuming a spacetime metric. A key result is that the Laplace operator emerges uniquely from locality, homogeneity, and isotropy of the lattice, closing the final logical gap in the MMU–Einstein correspondence.

1 Definitions and minimal assumptions

Definition 1 (UR–cell deformation field). Let $a(x)$ be the local microscopic edge length of the MMU ur–cell. Define the dimensionless scalar deformation

$$u(x) := \frac{a(x) - a_0}{a_0}, \tag{1}$$

with a_0 the asymptotic reference length.

Definition 2 (Internal coordinates and projection time). Each cell carries internal coordinates (w_1, w_2, w_3, w_4) and an internal evolution parameter σ. Observable time is defined by projection

$$dt := |\dot{w}_1|\, d\sigma, \tag{2}$$

while proper time is

$$d\tau := v_{\text{int}}\, d\sigma, \quad v_{\text{int}}^2 = \sum_{i=1}^{4} \dot{w}_i^2, \quad v_\perp^2 = \dot{w}_2^2 + \dot{w}_3^2 + \dot{w}_4^2. \tag{3}$$

Assumption A (Local quadratic dynamics). For small oscillations a single ur–cell is governed by

$$L = \frac{1}{2}\dot{w}^\mathsf{T} M \dot{w} - \frac{1}{2} w^\mathsf{T} K(a) w, \tag{4}$$

with $M > 0$ and stiffness scaling $K(a) \propto a^{-3}$.

Assumption B (Stationarity). A static gravitational configuration corresponds to adiabatic spatial variation of $a(x)$ with conserved internal energy along stationary worldlines.

2 Objective gravitational field

The deformation field $u(x)$ is a coordinate-independent scalar and constitutes the physical gravitational field of the MMU.

The field $u(x)$ is defined directly from the microscopic lattice length, which is a geometric property of the ur–cell. Its value is independent of coordinate parametrization and therefore represents a scalar field in the continuum limit.

3 Continuum limit and inevitability of the Laplace operator

3.1 Discrete lattice operator

Let u_i be the deformation at lattice site i. The most general local, linear, stationary interaction on a homogeneous lattice is

$$(\mathcal{L}u)_i = \sum_{j \in N(i)} c_{ij}\,(u_j - u_i), \tag{5}$$

where $N(i)$ denotes nearest neighbors.

3.2 Continuum expansion

Set $u_j = u(x + r_j)$ and expand

$$u(x + r_j) - u(x) = r_j \cdot \nabla u + \frac{1}{2} r_j^\mathsf{T} H(u)\, r_j + \mathcal{O}(|r_j|^3). \tag{6}$$

3.3 Symmetry constraints

Homogeneity implies constant coefficients $c_{ij} = c_j$. Isotropy implies that for each r_j there exists $-r_j$ with equal weight. Therefore

$$\sum_j c_j r_j = 0, \tag{7}$$

and all first-order terms vanish.

The second-order moment satisfies

$$\sum_j c_j\, r_j r_j^\mathsf{T} = \lambda I, \tag{8}$$

with I the identity matrix.

3.4 Emergent operator

Thus

$$(\mathcal{L}u)(x) = \frac{\lambda}{2}\,\mathrm{tr}\,H(u) = D\,\nabla^2 u. \tag{9}$$

Under locality, homogeneity, and isotropy, the Laplace operator is the unique leading-order continuum operator for the scalar deformation field.

All lower-order terms vanish by symmetry, and any anisotropic second-order operator is excluded by isotropy. Hence ∇^2 is unavoidable.

4 Poisson equation and exterior solution

The weak-field deformation obeys

$$\nabla^2 u(x) = \frac{4\pi G}{c^2}\rho_{\text{eff}}(x). \tag{10}$$

Dimensional consistency fixes the coupling to energy density, while the operator structure follows from the previous section.

Outside a spherical source,

$$u(r) = -\frac{GM}{rc^2}. \tag{11}$$

5 Energy closure and Schwarzschild redshift

In a static field,

$$\frac{v_\perp^2}{2} = -\Phi(r), \qquad \Phi(r) = c^2 u(r). \tag{12}$$

Conservation of internal energy requires elastic compression to be compensated by transverse kinetic energy.

The MMU projection factor satisfies

$$p(r) = \frac{w_1(r)}{w_1(\infty)} = \sqrt{1 - \frac{2GM}{rc^2}}, \tag{13}$$

and thus

$$g_{00}(r) = 1 - \frac{2GM}{rc^2}. \tag{14}$$

6 Geodesic motion

Define the line element

$$ds^2 = c^2 p(r)^2 dt^2 - d\ell^2. \tag{15}$$

Free motion extremises $\int ds$ and follows geodesics of the effective metric.

Proper time is the intrinsic evolution parameter. Stationary action implies extremal proper time, yielding the geodesic equation.

7 Conclusion

We have shown that (i) the gravitational field in the MMU is an objective scalar, (ii) the Laplace operator emerges uniquely in the continuum limit, (iii) the Poisson equation and Schwarzschild redshift follow inevitably, and (iv) free-fall is necessarily geodesic. Einstein gravity in the weak-field regime is therefore not an assumption, but a structural consequence of the MMU framework.